

HXC-108_android_evidence

HXC-108 — Android Client Recording Evidence (§11.4.158 / §11.4.143)

Run-id: HXC-108_android · **Date:** 2026-06-23 **Gate host (§11.4.119 single-owner):** thinker.local — Linux 6.17 x86_64, 16 cores, /dev/kvm present, podman 4.9.3 (rootless, §11.4.161). **Sanctioned path (§6.X):** helix_code/scripts/run-challenge-matrix.sh → Android emulator INSIDE a rootless podman container (image localhost/lava-android-emulator:api34-x86_64); install/launch/screenrecord driven in-container via podman exec / adb -e / podman cp. No host-direct emulator/adb. **Recording prefix (§11.4.155):** helixcode. **Window scope (§11.4.154):** the emulator device frame (1080×1920), not a desktop.

Durable evidence committed under docs/qa/HXC-108_android/ per §11.4.83 (NOT the rotatable git-ignored raw corpus — the exact §11.4.154-rotation bluff the HXC-107 audit caught; see docs/qa/HXC-107_ledger_audit.md).

Durable artifacts (committed)

File	What
docs/qa/HXC-108_android/helixcode-android-client.mp4	The recording — H.264, 1080×1920, 33 frames (ffprobe-verified).
docs/qa/HXC-108_android/helixcode-android-MainActivity.png	Key frame — the genuine HelixCode Android client.

Raw run artifacts (also on disk, git-ignored §11.4.128): /Volumes/T7/Downloads/Recordings/helixcode-android-client-20260623-190817.{mp4,png,logcat.txt}; on thinker ~/helixcode-recordings/.

Content verification (§11.4.158 / §11.4.107 / §11.4.123) — PASS

The recorded frame shows the **real HelixCode Android client MainActivity** running on Android 14 — NOT a black / boot-animation / crash / stub frame: - Green “**HelixCode**” title bar (brand). - The HelixCode spiral logo rendered. - Status “**Disconnected**” in red — a REAL runtime connection state, not placeholder text. - A “**User:**” label. - A green **CONNECT** button. - Android 14 system UI chrome (status bar clock 4:09, wifi/signal/battery icons).

Corroborating runtime signal (§11.4.108): dumsys reported ResumedActivity: dev.helix.code/.MainActivity (the app is genuinely foregrounded). APK install reported Success; the installed APK md5 (e85b19af9cdd292b6150bffab2dc277c) matched the macOS-built app-debug.apk (real native Go core embedded).

The conductor independently re-verified: `ffprobe` confirms h264 1080×1920 / 33 frames, and visually read the committed PNG (the description above is what is genuinely on screen).

Blockers hit + resolved during the run (FACTs, §11.4.6)

1. **Rootless /dev/kvm denial** — ProbeKVM failed (This user doesn't have permissions to use KVM): image runs as USER emulator (uid 1000) → host subuid lacks the /dev/kvm ACL. Fixed with `--userns=keep-id` (container-uid-1000 → host user who holds the ACL). Verified Boot completed in 29201 ms.
2. `podman cp $CN:/sdcard/...` “no such file” — /sdcard is in the emulator GUEST, not the container fs. Fixed with a two-hop pull: `adb -e pull guest→container /tmp`, then `podman cp container→host`.
3. **Intermittent early container exit** — emulator (PID 1) died if a screenshot ran first. Fixed by record-first ordering. No leftover `helixcode-emu` containers after teardown.

Submodule gap surfaced (§11.4.74 extend-don't-reimplement) — follow-up

`submodules/containers/pkg/emulator/containerized.go`
`buildContainerRunArgs` passes only `--device /dev/kvm` and lacks `--userns=keep-id`, so its `Containerized.Boot` would hit the same rootless ProbeKVM denial. **Recommended upstream fix:** add `--userns=keep-id` (rootless podman) in `buildContainerRunArgs`. The wrapper adds the flag itself for now and documents the gap inline; the submodule was NOT modified (left for conductor/operator decision per §11.4.74).

Verdict

Android §11.4.158 client recording: **PASS** — real, content-verified, via the §6.X-sanctioned containerized path on a Linux KVM host. The macOS-podman path remains §11.4.112 structurally-impossible (`docs/research/android_emulator_podman_macos_20260623/feasibility.md`); this is the research's recommended path [A] realised.